

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY
THIRD SEMESTER – B.TECH.
END-SEMESTER THEORY EXAMINATION, JULY 2026

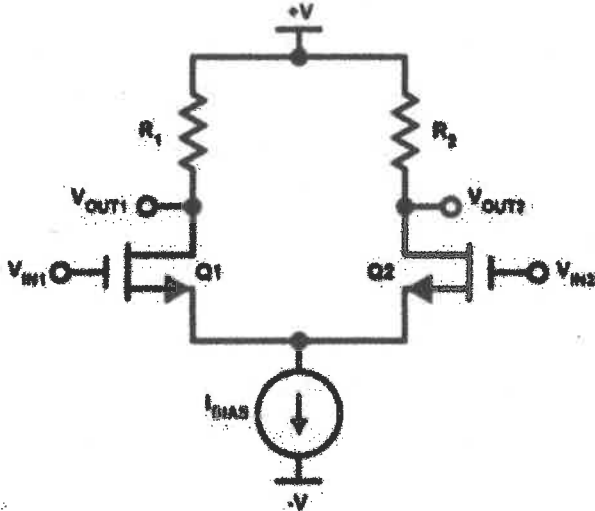
Course Code: ECECC304/EIECC07

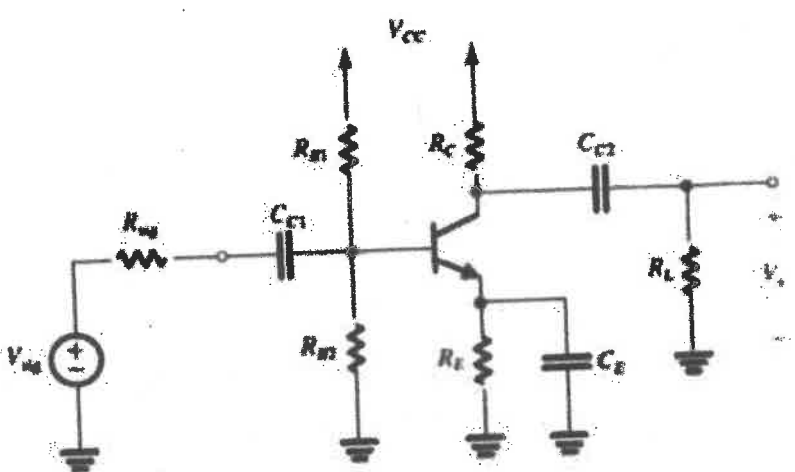
Course Title: Microelectronics Circuits and Application

Time: 3:00 Hours

Max. Marks: 50

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	A BJT differential amplifier is biased with a tail current of 2 mA. Each transistor has $\beta = 120$, $V_t = 25$ mV, and collector resistance $R_C = 6.8$ k Ω . (i) Calculate the differential gain. (ii) Determine the output voltage for a differential input of 8 mV.	5	CO1
1b	A Wilson current mirror has $I_{REF} = 500\mu A$, $\beta = 150$, Determine, (i) output current, (ii) output resistance	5	CO1
1c	For the MOS differential pair with a common-mode voltage V_{CM} applied, as shown in Fig. 1, let $V_{DD}=V_{SS}= 1.5$ V, $k_n(W/L) = 4\text{mA/V}^2$, $V_t = 0.5$ V, $I = 0.4$ mA, and $R_D = 2.5$ k Ω , and neglect channel-length modulation. Assume that the current source I requires a minimum voltage of 0.4 V to operate properly. Find VOV and V_{GS} for each transistor.  Fig. 1.	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	Derive the voltage gain of a Cascode Amplifier and compare its gain and bandwidth with a Common Source amplifier.	5	CO2
2b	A common-emitter amplifier has voltage gain $A_v = -120$, collector-base capacitance, $C_{\mu} = 2.5\text{pF}$, base-emitter capacitance, $C_{\pi} = 18\text{pF}$, source resistance, $R_S = 1.5\text{k}\Omega$, input resistance of transistor, $r_{\pi} = 3.5\text{k}\Omega$. Using Miller's theorem: (i) Calculate the effective input capacitance, (ii) Determine the upper 3-dB cutoff frequency considering only the input pole.	5	CO2

2c	Using the Open Circuit Time Constant (OCTC) method, derive the expression for the upper cut-off frequency the common-emitter BJT amplifier shown in Fig. 2. Clearly state all assumptions made during the derivation.  Fig. 2.	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	Explain the working principle of a negative feedback amplifier using the general feedback structure. Derive the gain equation.	5	CO3
3b	Derive the expressions for the input resistance and output resistance of a Series-Shunt (Voltage-Series) feedback amplifier. Explain the effect of negative feedback on both resistances.	5	CO3
3c	A voltage-series feedback amplifier has open-loop voltage gain $A_v = 250$, feedback factor $\beta = 0.04$, input resistance $R_i = 3 \text{ k}\Omega$, output resistance $R_o = 60 \text{ k}\Omega$. Calculate: (i) Loop gain, (ii) Closed-loop voltage gain, (iii) Input resistance with feedback, and (iv) Output resistance with feedback.	5	CO3
Q 4	Attempt any 2 parts of the following		
4a	Draw the circuit of a Class A power amplifier. Explain its operation with transfer characteristics and derive maximum efficiency.	5	CO4
4b	A Class B amplifier has $V_{CC} = 18\text{V}$, $R_L = 8 \text{ }\Omega$. Determine, (i) Maximum output power, (ii) DC input power, (iii) Maximum efficiency.	5	CO4
4c	Explain the working of a push-pull Class B amplifier with a neat circuit diagram. State its advantages over a single-ended Class B amplifier.	5	CO4
Q 5	Attempt any 2 parts of the following		
5a	Draw the internal block diagram of the 555-timer IC. Explain the function of each internal block.	5	CO5
5b	A 555 astable multivibrator has $R_A = 12 \text{ k}\Omega$, $R_B = 18 \text{ k}\Omega$, $C = 0.01 \text{ }\mu\text{F}$, Calculate (i) T_{on} , (ii) T_{off} , (iii) Time period, (iv) Frequency, and (v) Duty cycle.	5	CO5
5c	Draw the circuit of an Op-Amp RC phase shift oscillator. Derive the expression for the frequency of oscillation and gain condition.	5	CO5